PARTICULATE NUTRIENT DYNAMICS IN THE GREAT LAKES

Grad 505 Final Project
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Project Overview

Data source: EPA Great Lakes National Program Office dataset

Years analyzed: 2006–2019

Nutrients studied:

- Particulate Total Nitrogen (PTN)
- Particulate Total Phosphorus (PTP)
- Particulate Organic Carbon (POC)
- Total Suspended Solids (TSS)

Goal: understand spatial, temporal, and seasonal nutrient patterns in the Great Lakes

Research Questions

- Have particulate nutrients in **Lake Michigan changed over time?**
- Can nutrient measurements **predict which lake a sample came from?**
- Do **outliers represent unusual environmental conditions**

Methods

EDA and Statistical Analysis

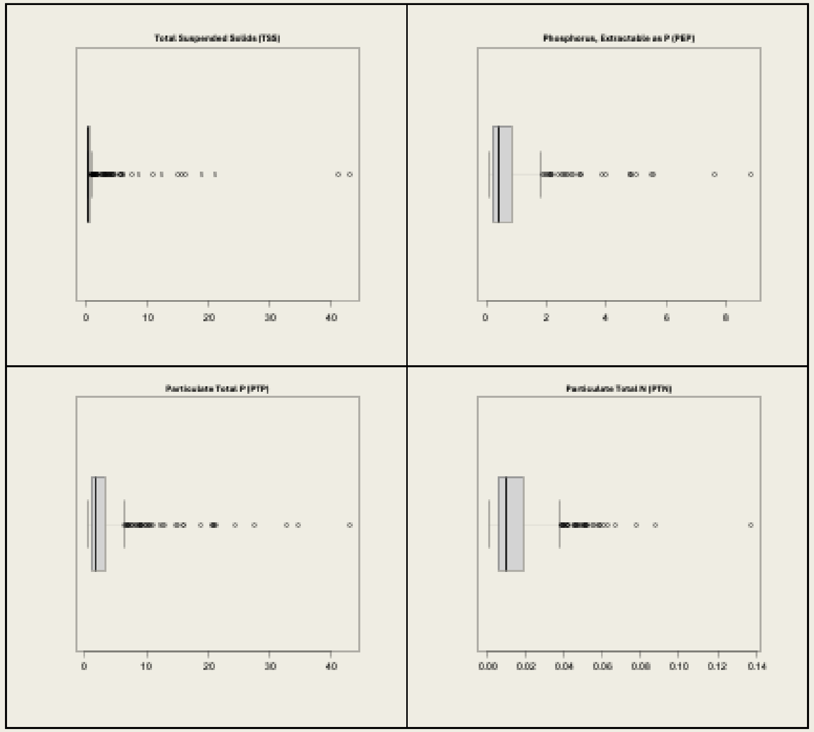
EDA showed:

- Strong **right skew** in nutrient variables
- Several **upper outliers**
- Log transformations required before modeling

These observations guided the modeling approach.

Statistical Analysis included:

- Log transformed linear regression
- Multinomial classification models
- Permutation testing
- Logistic regression for outliers



This visual shows the distributions of each particulate nutrient

Time Trends in Lake Michigan

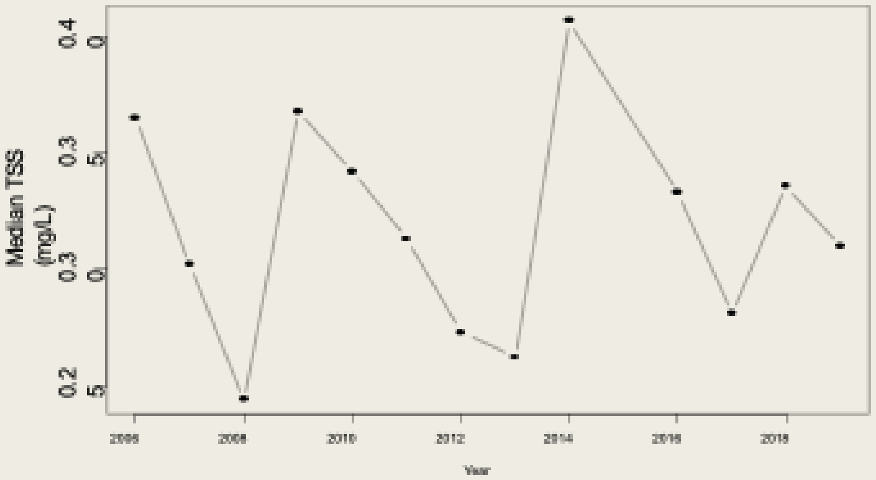
KEY RESULTS RQ1

Nitrogen (PTN) showed a significant trend over time

No significant change detected for

- *Phosphorus (PTP)*
- *Organic Carbon (POC)*
- *Suspended Solids (TSS)*

Implication: most particulate nutrients remained stable

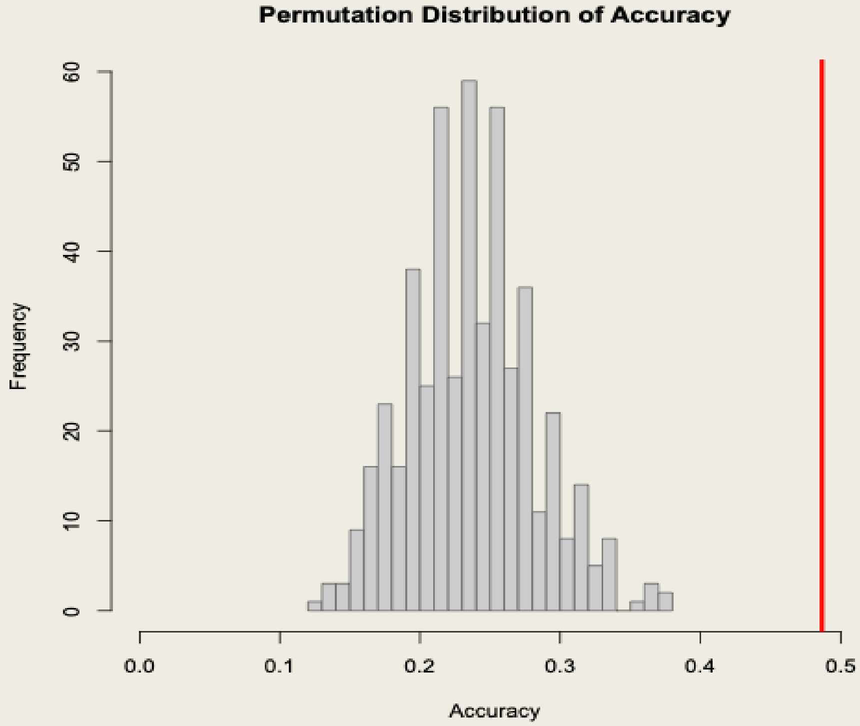


This visual shows variability over time, yet lacks strong monotonic trend

Can Nutrients Predict Lake Identity

KEY RESULTS RQ2

- Baseline model accuracy: ~23%
- Season only model: ~22%
- *Full nutrient model*: ~49%
- Permutation test p-value = 0.001



This visual shows permutation distribution + lake prediction accuracy by model

Outliers and Environmental Conditions

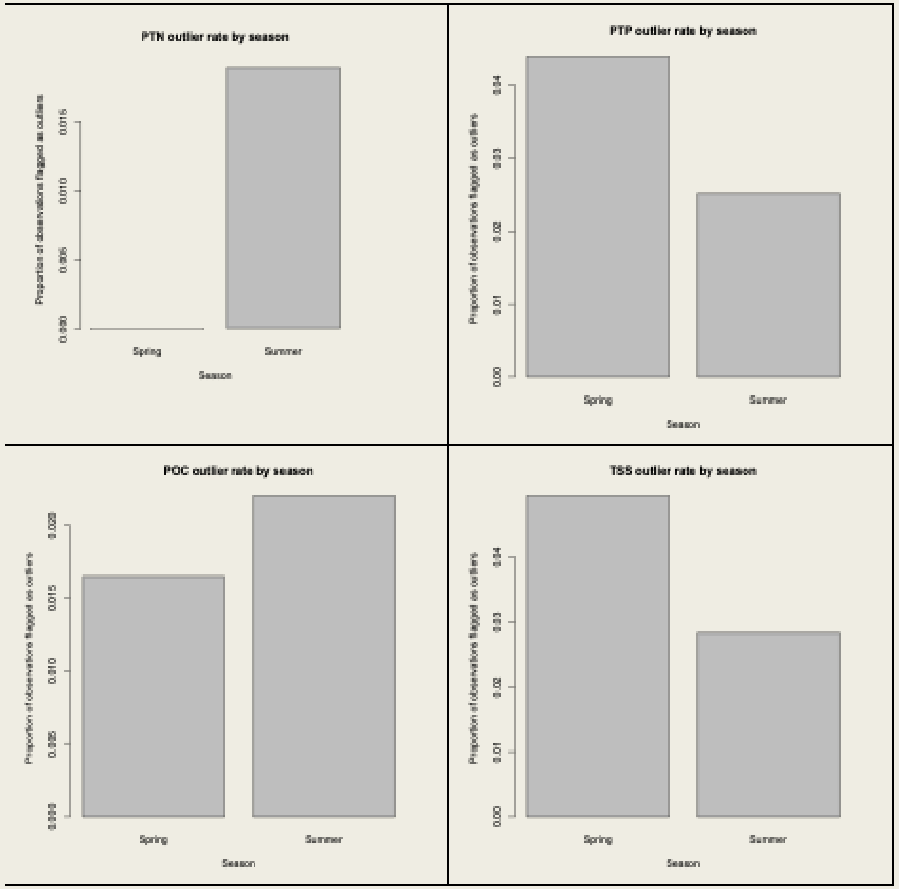
KEY RESULTS RQ3

Outliers defined using the 1.5×IQR rule

Logistic regression tested seasonal influence

Results:

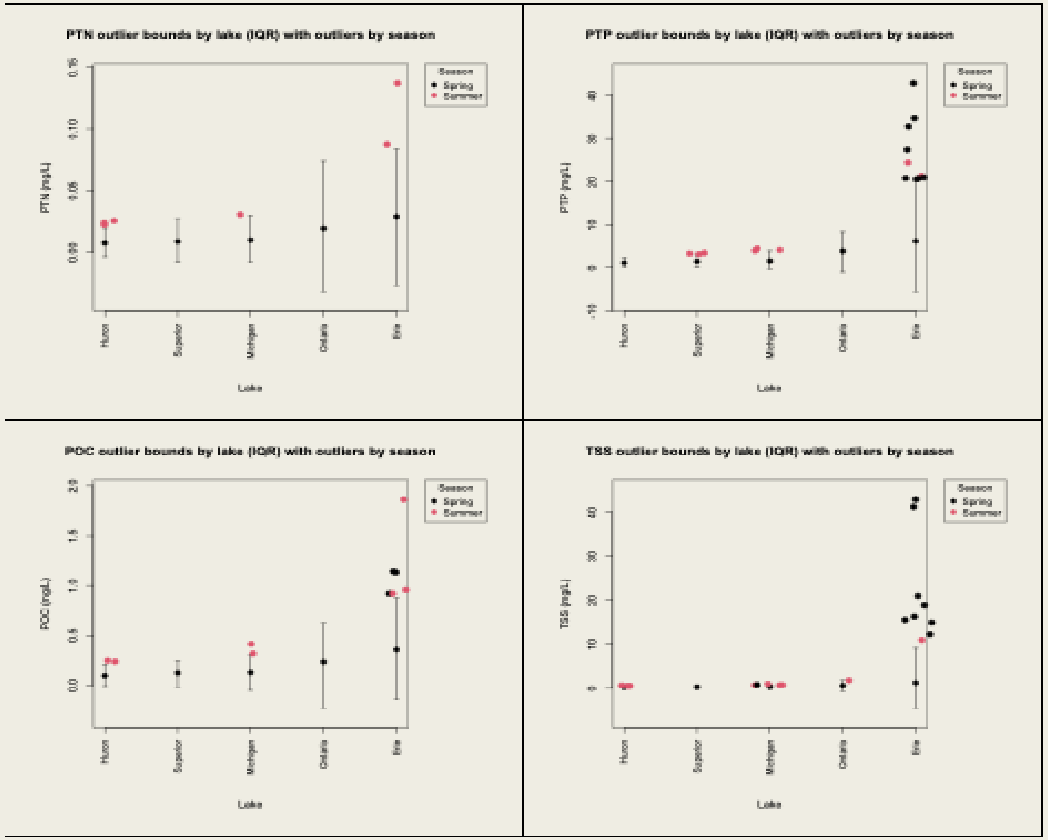
- Season predicts PTN outliers
- Season does not predict PTP, POC, or TSS outliers



This visual shows differences by nutrient in seasonal proportions

Key Conclusions

- Great Lakes show distinct particulate nutrient signatures
- Most nutrients are stable over time in Lake Michigan
- Nitrogen shows evidence of change and seasonal anomalies
- Environmental differences between lakes remain measurable



This visual reinforces lake differences in nutrient concentration by season

THE END